

Ellen Su

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Education

New York University

2024 - present

- Ph.D. Student, Center for Data Science
- Advisor: Todd Gureckis

Princeton University

2019 - 2023

- B.S. Computer Science
- Advisor: Tom Griffiths
- Thesis: *Revealing the Priors of Deep Learning Models Through Iterated Learning*

Publications

Su, E.*, Potapczynski, A.*, Qiu, S., Hughes, E. & Gordon Wilson, A. (2026). Epiplexity Guided Data Selection and Generation for Out-of-Distribution Generalization. <https://arxiv.org/abs/2608.11746>.

Su, E.*, Legris, S.*, Gureckis, T.M., & Ren, M. (2025). Opinion: Learning Intuitive Physics Requires More than Visual Data. *NeurIPS Embodied World Models for Decision Making Workshop 2025*.

Su, E., Ho, M.K., & Gureckis, T.M. (2025). Integration of Language and Experience via the Instructed Bandit Task. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society, CogSci 2025*.

Su, E., Vellore, A., Chang, A., Mura, R., Nelson, B., Kassianik, P., & Karbasi, A. (2024). Extracting Memorized Training Data via Decomposition. <https://arxiv.org/abs/2409.12367>.

Arevalo, J.*, Su, E.*, Carpenter, A., & Singh, S. (2024). MOTIVE: A Drug-Target Interaction Graph For Inductive Link Prediction. *NeurIPS Datasets and Benchmarks. (spotlight)*

Arevalo, J., Su, E., Ewald, J.D., Van Dijk, R., Carpenter, A., & Singh, S. (2024). Evaluating batch correction methods for image-based cell profiling. *Nature Communications, 15*.

* *equal contribution*

Awards / Invited Talks

2026	Meta AI Mentorship (AIM) Program Fellowship
2026	Brains, Minds, & Machines NIH Scholarship
2025	Computation and Decision-Making Lab, NYU
2025	Human and Machine Learning Lab, NYU
2024	Spotlight, NeurIPS 2024 (top 2.08% of submissions) <i>MOTIVE: A Drug-Target Interaction Graph For Inductive Link Prediction.</i>
2023	Princeton CS Outstanding Student Teaching Award
2022	Princeton CS Grace Hopper Celebration Grant
2021	Princeton Summer Research Fellowship

Research

Meta Fundamental AI Research (FAIR)

New York, NY

Visiting Researcher, Mentors: Florian Bordes and Adina Williams

Sept 2026 - Present

- Advancing physical reasoning in video models

Robust Intelligence

San Francisco, CA

Research Intern, PI: Amin Karbasi

Summer 2024

- Developed a decomposition jailbreak to extract training data from production LLMs

Broad Institute of MIT and Harvard

Cambridge, MA

ML Research Associate, PIs: Anne Carpenter, Shantanu Singh

2023 - 2024

- Used graph neural networks to advance drug-target interaction (DTI) discovery
- Produced a co-first author spotlight publication in NeurIPS and second author publication in Nature Communications

Princeton University

Princeton, NJ

Senior Thesis, PI: Tom Griffiths

2022 - 2023

- Used convolutional neural networks (CNNs) in iterated learning chains to identify inductive biases
- Demonstrated that machine priors manifested in sequential decision-making behavior
- Presented at the Program in Applied and Computational Mathematics Symposium

Junior Paper, PI: Ben Raphael

2021 - 2022

- Developed to a machine learning algorithm that predicts copy number variation mutations from spatial transcriptomics data

Teaching

DS-GA1012: LLMs: Evaluations and Applications

NYU, NY

Graduate Course TA & Guest Lecturer

Spring 2026

Princeton Tutoring

Princeton, NJ

Math/Physics/CS Tutor

2021 - 2024

COS495: Web3/Blockchains

Princeton, NJ

Undergraduate Course Assistant

Spring 2023

Reviewing

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- Neurips 2026
 - Cognitive Computational Neuroscience 2026
 - Cognitive Science 2025

Service

Princeton First Aid and Rescue Squad

Princeton, NJ

National Registry of Emergency Medical Technicians

2021 - 2023

Skills and Languages

Interests

Machine learning, cognitive science, artificial intelligence

Languages (program)	Python (PyTorch, JAX), Java, JavaScript, C, SQL, CSS
Languages	English (native), Mandarin (native), Spanish (advanced)